

Answer to the Q. NO - 1(a)

Given, $K' = 0.2 \text{ mA/V}^2$; $V_t = 0.7 \text{ V}$; $\frac{W}{L} = 10$

(a) $V_{GS} = 5 \text{ V}$; $V_{DS} = 1 \text{ V}$

Here, $V_{GS} > V_t$

and $V_{GS} - V_t = 5 - 0.7 = 4.3 \text{ V}$

$\therefore V_{DS} < V_{GS} - V_t$ [Triode mode]

$\therefore I_D = K' \frac{W}{L} \left(V_{GS} - V_t - \frac{1}{2} V_{DS} \right) V_{DS}$

$= 0.2 \times 10 \times \left(4.3 - \frac{1}{2} \right)$

$= 7.6 \text{ mA}$ (Ans)

(b) $V_{GS} = 2$, $V_{DS} = 1.3 \text{ V}$

Here, $V_{GS} > V_t$

and $V_{GS} - V_t = 2 - 0.7 = 1.3$

$\therefore V_{DS} \geq V_{GS} - V_t$ [saturation mode]

$\therefore I_D = \frac{K'}{2} \frac{W}{L} (V_{GS} - V_t)^2$

$= \frac{0.2}{2} \times 10 \times (1.3)^2 = 1.69 \text{ mA}$ (Ans)

c) $V_{GS} = 5V$, $V_{DS} = 0.2V$ (Answer to the question)

Here, $V_{GS} > V_t$; $V_{GS} = 5V$, $V_t = 0.7V$ (given) (1)

and $V_{GS} - V_t = 5 - 0.7 = 4.3V$

$\therefore V_{DS} < V_{GS} - V_t$ [Triode mode] $V_{DS} = 0.2V$ (2)

$\therefore I_D = K' \frac{W}{L} \left(V_{GS} - V_t - \frac{1}{2} V_{DS} \right) V_{DS}$ (Here $V_{GS} > V_t$)

$= 0.2 \times 10 \times \left(4.3 - \frac{0.2}{2} \right) \times 0.2$ (3)

$= 1.68 \text{ mA}$ [Ans] (4)

d) $V_{GS} = V_{DS} = 5V$ (5)

Here, $V_{GS} > V_t$

and $V_{GS} - V_t = 4.3$

$\therefore V_{DS} \geq V_{GS} - V_t$

$\therefore I_D = \frac{K'}{2} \frac{W}{L} (V_{GS} - V_t)^2$ (6)

$= \frac{0.2}{2} \times 10 \times (4.3)^2 = 18.49 \text{ mA}$ (Ans) (7)

[Ans] (8)

$I_D = \frac{K'}{2} \frac{W}{L} (V_{GS} - V_t)^2$

(Ans) (9)

Answer to the Q. NO - 1(b)

$$V_{th} = 1V$$

case	V_{GS}	V_{OV}	V_{DS}	Region of operation
a	0	-1	1	cutoff ; $V_{GS} < V_{th}$
b	1.5	0.5	1	saturation ; $V_{DS} > V_{th}$ $V_{DS} \geq V_{OV}$
c	1.5	0.5	0.5	saturation ; $V_{DS} \geq V_{OV}$
d Here, $V_D = 1V$ $V_S = 0V$	1.5	0.5	1	saturation ;
e	2.5	1.5	1-1 = 0	Triode ; $V_{GS} > V_{th}$ $V_{DS} < V_{OV}$

Answer to the Q. NO - 1(c)

1st circuit,

$$\therefore V_{GS} = 2.2 - 0 = 2.2V$$

$$V_G = 2.2V$$

$$V_{OV} = 2.2 - 0.4 = 1.8V$$

$$V_S = 0V$$

$$V_{DS} = 2.2 - 0 = 2.2V$$

$$V_D = 2.2V$$

$$V_T = 0.4$$

$$\therefore V_{DS} \geq V_{OV}$$

$\therefore$ Region of operation is saturation. ~~made~~

2nd circuit,

(d) $\downarrow$ - ON. 2 out of 4 answers
 $\therefore V_{GS} = 0 - (-1) = 1V$

$V_G = 0V$

$V_{OV} = 1 - 0.4 = 0.6V$

$V_S = -1V$

$V_{DS} = -0.6 - (-1) = 0.4V$

$V_D = -0.6V$

$V_T = 0.4V$

$V_{DS} < V_{OV}$

$\therefore$ Region of operation is triode.

3rd circuit,

$V_G = 1V$

$\therefore V_{GS} = 1 - 1 = 0V$

$V_S = 1V$

$V_D = 3V$

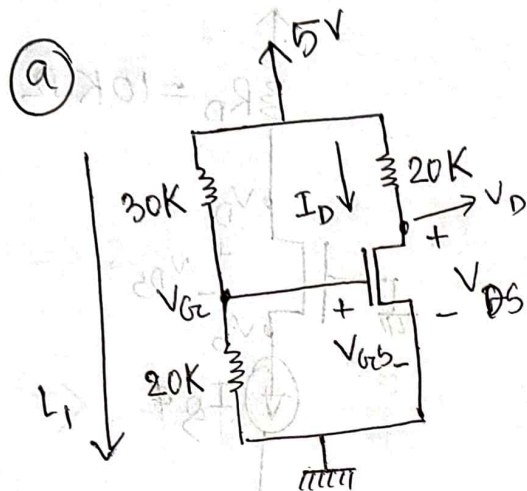
$V_T = 0.4V$

$\therefore V_{GS} < V_T$

$\therefore$ Region of operation is cutoff.

$V_{OV} \leq 0V$

Answers to the Q. NO-2



Here, $V_{DS} = 5V$ (assumed)

$$V_{th} = 1V$$

$$K_n = 0.1 \text{ mA/V}^2$$

$$\text{and } V_S = 0V$$

KVL in L_1 ,

$$30I + 20I = 5 \Rightarrow 50I = 5 \Rightarrow I = 0.1 \text{ mA}$$

$$\text{Now, } \frac{V_G - 0}{20} = 0.1$$

$$\Rightarrow V_G = 2V$$

Easier to do voltage division:

$$V_G = \frac{20K}{20K + 30K} \times 5 = 2V$$

$$\therefore V_{GS} = 2 - 0 = 2V > V_{th}$$

Let's assume ~~active~~ saturation mode,

$$\therefore I_D = \frac{K_n}{2} (V_{GS} - V_{th})^2$$

$$\Rightarrow I_D = \frac{0.1}{2} \times (2 - 1)^2$$

$$\Rightarrow I_D = 0.05 \text{ mA}$$

At V_D node,

$$\frac{5 - V_D}{20} = 0.05$$

$$\Rightarrow V_D = 4V$$

$$\therefore V_{DS} = 4 - 0 = 4V$$

$$V_{ov} = 2 - 1 = 1V$$

$$\therefore V_{DS} > V_{ov}$$

Assumption correct!

Answer to the Q. NO - 3

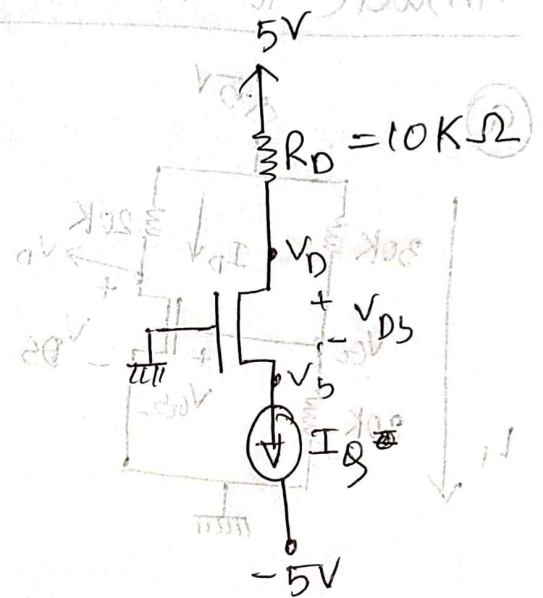
(a) Here, $V_D = 2.5 \text{ V}$

$$V_{Th} = 0.8 \text{ V}$$

$$K'_n = 80 \text{ MA/V}^2$$

$$V_{G1} = 0 \text{ V}$$

$$V_{G2} = 5 \text{ V}$$



$$V_D = 2.5 \text{ V} > V_{G1}$$

$\therefore$ The mosfet is in saturation.

(V_S not needed)

~~let's assume~~ saturation,

$$I_D = \frac{K_n}{2} \frac{W}{L} (V_{GS} - V_{th})^2$$

$$\Rightarrow 250 = \frac{80}{2} \frac{W}{L} (-V_S - 0.8)^2$$

$$\Rightarrow \frac{W}{L} = \frac{25}{4} \times \frac{1}{V_S^2 + 1.6V_S + 0.64}$$

(b) Here,

$$V_{thL} = V_{thD} = 1V$$

$$K_{nD} = 50 \text{ Ma/V}^2$$

$$K_{nL} = 10 \text{ Ma/V}^2$$

$$V_{DD} = 5V$$

→ No need to assume.
we know this.

let's assume M_L in saturation and M_D in triode,

$$\text{Here, } I_{DL} = I_{DD}$$

$$\Rightarrow \frac{K_{nL}}{2} (V_{GS} - V_{thL})^2 = K_{nD} \left(V_{GSD} - V_{thD} - \frac{1}{2} V_{DSD} \right) V_{DSD}$$

$$\Rightarrow \frac{10}{2} (5 - V_0 - 1)^2 = 50 \times \left(5 - 0 - 1 - \frac{1}{2} V_0 \right) V_0$$

$$\Rightarrow 5 \times (4 - V_0)^2 = 50 \times \left(4 - \frac{1}{2} V_0 \right) V_0$$



$$\Rightarrow 16 - 8V_0 + V_0^2 = 40V_0 - 5V_0^2$$

$$\Rightarrow 6V_0^2 - 48V_0 + 16 = 0$$

$$\therefore V_0 = \frac{7.65, 0.35}{x}$$

$$\therefore V_{GS_L} = 5 - V_0 = 5 - 0.35 = 4.65 \text{ V}$$

$$V_{DS_L} = 5 - V_0 = 5 - 0.35 = 4.65 \text{ V}$$

$$V_{OVL} = 4.65 - 1 = 3.65 \text{ V}$$

$$\therefore V_{DS_L} \geq V_{OVL} \text{ [saturation mode]}$$

$\therefore$ assumption correct!

$$V_{GS_D} = 5 - 0 = 5 \text{ V}$$

$$V_{DS_D} = 0.35 - 0 = 0.35 \text{ V}$$

$$V_{OVD} = 5 - 1 = 4 \text{ V}$$

$$\therefore V_{DS_D} < V_{OVD} \text{ [triode mode]}$$

$\therefore$ assumption correct!

(e) Given,

$$V_{ov} = 5 - 0.6 = 4.4 \text{ V}$$

$$I_D = 12 \text{ mA}$$

$$V_{GS} = 5 \text{ V}$$

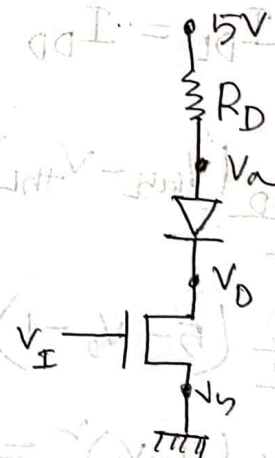
$$V_S = 0 \text{ V}$$

$$V_{GS} = 5 - 0 = 5 \text{ V}$$

$$V_{DS} = 0.15 \text{ V}$$

$$V_{Th} = 0.6 \text{ V}$$

$\therefore$ Region of operation is triode.



$$I_D = K'_n \frac{W}{L} \left(V_{GS} - V_{Th} - \frac{1}{2} V_{DS} \right) V_{DS}$$

$$\Rightarrow \frac{W}{L} = \frac{I_D}{K'_n \left(V_{OV} - \frac{1}{2} V_{DS} \right) V_{DS}}$$

$$\Rightarrow \frac{W}{L} = \frac{12 \times 10^{-3}}{80 \times 10^{-6} \times \left(4.4 - \frac{0.15}{2} \right) 0.15}$$

$$\Rightarrow \frac{W}{L} = 231.21 \approx 231 \quad (\text{Ans})$$

Now, $V_a - V_D = 1.6 \text{ V}$

$$\Rightarrow V_a = 1.6 + 0.15$$

$$\Rightarrow V_a = 1.75$$

$$\begin{aligned} V_{DS} &= V_D - V_S \\ \Rightarrow 0.15 &= V_D - 0 \\ \Rightarrow V_D &= 0.15 \end{aligned}$$

$$\therefore \frac{5 - V_a}{R_D} = I_D$$

$$\Rightarrow \frac{5 - 1.75}{R_D} = 12 \times 10^{-3}$$

$$\Rightarrow R_D = 270.83 \, \Omega \approx 271 \, \Omega$$

$$(\text{Ans})$$

Answer to the Q. NO-4

Let's $V_{CE(sat)} = 0.2 \text{ V}$ [For active mode]

Case-1°

$$V_{BE} = 0.7 \text{ V}$$

$$V_{CE} = 0.7 \text{ V}$$

$$\therefore V_{CE} > V_{CE(sat)}$$

$\therefore$ Mode of operation is Active.

Case-2°

$$V_{BE} = 0.8 \text{ V}$$

$$V_{CE} = 0.1 \text{ V}$$

$\therefore$ Mode of operation is Saturation.

Case-3°

$$V_{BE} = 0.7 \text{ V}$$

$$V_{CE} = 1.4 \text{ V}$$

$\therefore$ Mode of operation is Active.

$$\therefore V_{CE} > V_{CE(sat)}$$

Case-4°

$$V_{BE} = 0.7 \text{ V}$$

$$V_{CE} = 2.7 \text{ V}$$

$\therefore$ Mode of operation is Active.

$$\therefore V_{CE} > V_{CE(sat)}$$

Answer to the Q. NO-5

(a) Here, $V_{BE} = 0.8 \text{ V}$, $V_C = 0.5 \text{ V}$

$\beta = 100$

$V_B = 0 \text{ V}$

$\therefore V_E = -0.8 \text{ V}$ $\therefore V_{CE} = 0.5 + 0.8$
 $= 1.3 > V_{CE(sat)}$

mode is
 $\therefore$ Operation in active mode,

$I_C = 1 \text{ mA}$

$I_C = \beta I_B$

$\Rightarrow I_B = \frac{1}{100} = 0.01 \text{ mA}$

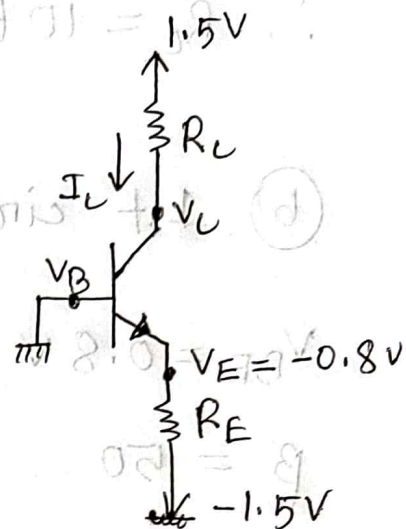
Now, $I_C = 0.1 \text{ mA}$

$\frac{1.5 - V_C}{R_C} = I_C$

$\Rightarrow R_C = \frac{1.5 - 0.5}{0.1} = 10 \text{ K}\Omega$

Again, $\frac{V_E - (-1.5)}{R_E} = I_E$

$\Rightarrow R_E = \frac{-0.8 + 1.5}{0.11} = 6.36 \text{ K}\Omega$



$$\therefore R_C = 10 \text{ K}\Omega \text{ and } R_E = 6.36 \text{ K}\Omega \text{ (Ans)}$$

(b) 1st circuit,

$$V_{BE} = 0.8 \text{ V}$$

$$\beta = 50$$

$$\text{Here, } V_B = 1.5 \text{ V}$$

$$\begin{aligned} \therefore V_E &= V_B - V_{BE} \\ &= 0.7 \text{ V} \end{aligned}$$

Let's assume active mode,

$$\therefore V_{BE} = 0.7$$

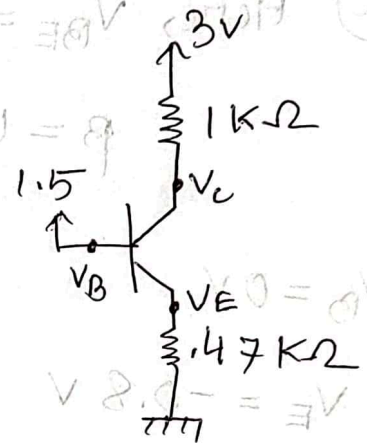
$$\text{and } V_{CE} > 0.2$$

At V_E Node,

$$\frac{V_E - 0}{0.47} = I_E \Rightarrow I_E = \frac{0.7}{0.47} \Rightarrow I_E = 1.489 \text{ mA}$$

$$I_C = \beta \alpha I_E \Rightarrow I_C = \frac{\beta}{\beta + 1} I_E \Rightarrow I_C = \frac{50}{50 + 1} \times 1.489 = 1.46 \text{ mA}$$

$$\begin{aligned} \therefore I_B &= I_E - I_C \\ &= 1.489 - 1.46 \\ &= 0.03 \text{ mA} \end{aligned}$$



At V_C node,

$$\frac{\beta - V_C}{1} = 1.46$$

$$\Rightarrow V_C = 1.54 \text{ V}$$

$$\therefore V_{CE} = 1.54 - 0.7 = 0.84 > 0.2$$

$\therefore$ assumption correct!

$$V_E = 0.7 \text{ V}$$

$$V_B = 1.5 \text{ V}$$

$$V_C = 1.54 \text{ V}$$

$$I_E = 1.489 \text{ mA}$$

$$I_B = 0.03 \text{ mA}$$

$$I_C = 1.46 \text{ mA}$$

(Ans)

2nd circuit,

let's assume active mode,

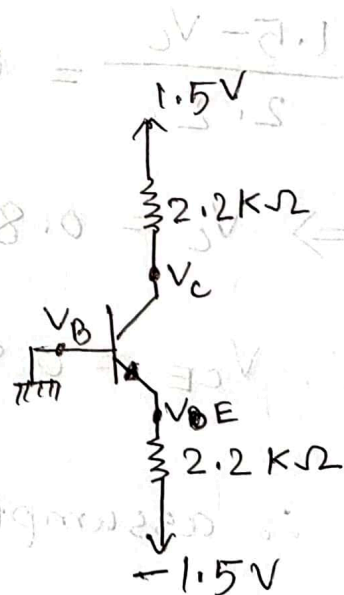
$$V_{BE} = 0.8 \text{ V}$$

$$\beta = 50$$

$$V_{CE} > 0.2$$

$$V_B = 0 \text{ V}$$

$$\therefore V_E = -0.8 \text{ V}$$



$$\therefore I_E = \frac{V_E - (-1.5)}{2.2}$$

$$\Rightarrow I_E = \frac{-0.8 + 1.5}{2.2} = 0.318 \text{ mA}$$

$$I_C = \frac{\beta}{\beta + 1} \times I_E$$

$$= \frac{50}{50 + 1} \times 0.318 = 0.312 \text{ mA}$$

$$I_B = I_E - I_C = 0.318 - 0.312$$

$$= 0.006 \text{ mA}$$

At V_C node,

$$\frac{1.5 - V_C}{2.2} = 0.318$$

$$\Rightarrow V_C = 0.8$$

$$\therefore V_{CE} = 0.8 - (-0.8) = 1.6 > 0.2$$

$\therefore$ assumption correct!

$$V_E = -0.8 \text{ V}$$

$$V_B = 0 \text{ V}$$

$$V_C = 0.8 \text{ V}$$

$$I_E = 0.318 \text{ mA}$$

$$I_B = 0.006 \text{ mA}$$

$$I_C = 0.312 \text{ mA}$$

(Ans)

Answer to the Q.NO - 6

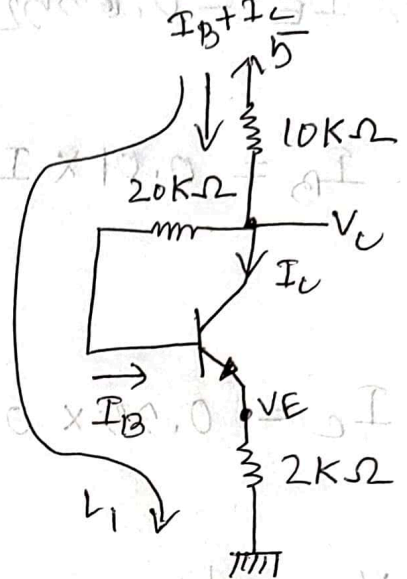
(a) Circuit-1:

Let's assume active mode,

$$V_{BE} = 0.7 \text{ V}$$

$$V_{CE} > 0.2 \text{ V}$$

$$\beta = \text{75} \rightarrow 100$$



KVL in L_1 ,

$$10(I_B + I_C) + 20I_B + 2I_E = 5 - 0.7 \quad \text{--- (1)}$$

$$I_E = I_B + I_C \quad \text{--- (2)}$$

$$\text{and, } I_C = \frac{\beta}{\beta + 1} I_E = 0.99 I_E$$

Here $\beta = 100$ was used.

Putting $I_C = 0.99 I_E$ in eqn (2),

$$I_E = I_B + 0.99 I_E$$

$$\Rightarrow I_B = 0.01 I_E$$

Putting $I_B = 0.01 I_E$ in eqn (1),

$$10 I_E + 20 \times 0.01 I_E + 2 I_E = 4.3$$

$$\Rightarrow 12.2 I_E = 4.3$$

$$\Rightarrow I_E = 0.0352 \text{ mA}$$

$$\therefore I_B = 0.01 \times I_E = 0.01 \times 0.0352 \text{ mA}$$

$$= 0.00352 \text{ mA}$$

$$I_C = 0.99 \times 0.0352 = 0.348 \text{ mA}$$

At V_C node,

$$\frac{5 - V_C}{10} = I_C + I_B$$

$$\Rightarrow V_C = 5 - 10(0.00352 + 0.348)$$

$$\Rightarrow V_C = \boxed{1.48} \text{ V}$$

At V_E node,

$$\frac{V_E}{2} = I_E$$

$$\Rightarrow V_E = 0.704 \text{ V}$$

$$\therefore V_{CE} = \boxed{1.48} - 0.704 = \boxed{0.776} > 0.2$$

[Active mode]

$\therefore$ assumption correct!

$$\therefore V_C = \boxed{1.48} \text{ V}$$

$$I_B = 0.00352 \text{ mA} \quad (\text{Ans})$$

$$A_m \text{ F.V.F.} = \frac{0.00352 \times 1000}{0.00352} = 1$$

Circuit-2:

Let's assume active mode,

$$V_{BE} = 0.7 \text{ V}$$

$$V_{CE} > 0.2 \text{ V}$$

$$\beta = 75$$

KVL on L_1 ,

$$10I_B + 4I_E = 0 - 0.7 + 8$$

$$\Rightarrow 10I_B + 4I_E = 7.3 \quad \text{--- (i)}$$

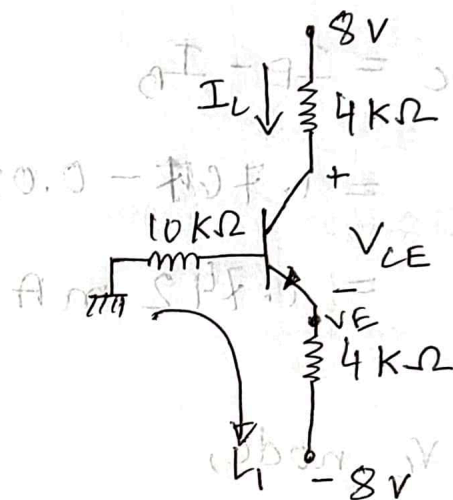
we know, $I_E = I_B + I_C$ and, $I_E = \frac{\beta}{\alpha} I_B$

$$= \frac{75}{0.99} I_B \quad \text{--- (ii)}$$

Putting eqn (ii) values on eqn (i),

$$10I_B + 4 \times \frac{75}{0.99} I_B = 7.3$$

$$\Rightarrow 313.03 I_B = 7.3$$



$$\frac{75}{76} = 0.986$$

$$\Rightarrow I_B = 0.0233 \text{ mA}$$

$$\therefore I_E = \frac{0.75}{0.99} \times 0.0233 = 1.765 \text{ mA}$$

and,

$$\begin{aligned} I_C &= I_E - I_B \\ &= 1.765 - 0.0233 \\ &= 1.742 \text{ mA} \end{aligned}$$

At V_C node,

$$\frac{8 - V_C}{4} = 1.742$$

$$\Rightarrow V_C = 1.032 \text{ V}$$

At V_E node,

$$\frac{V_E + 8}{4} = 1.765$$

$$\Rightarrow V_E = -0.94 \text{ V}$$

$$\therefore V_{CE} = 1.032 + 0.94 = 1.972 > 0.2$$

[Active mode]

$\therefore$ assumption correct!

$$\therefore I_C = 1.742 \text{ mA}$$

$$V_{CE} = 1.072 \text{ V}$$

(Ans)

(b) Let's assume active mode,

$$V_{BE} = 0.7 \text{ V}$$

$$V_{CE} > 0.2$$

$$\text{Here, } V_B = -1 \text{ V}$$

$$\therefore V_E = -1 - 0.7 = -1.7 \text{ V}$$

At V_E node,

$$\frac{V_E + 3}{4.8} = I_E$$

$$\Rightarrow I_E = \frac{-1.7 + 3}{4.8}$$

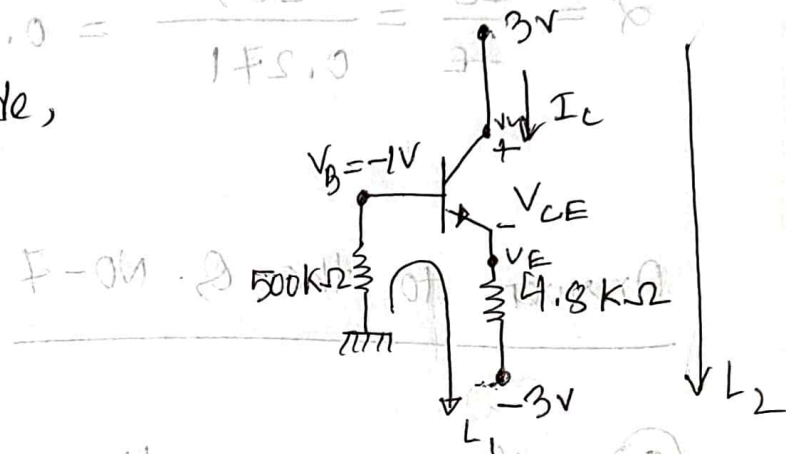
$$\Rightarrow I_E = 0.271 \text{ mA}$$

Now, KVL on L_1 ,

$$500 I_B + 4.8 I_E = -0.7 + 3$$

$$\Rightarrow 500 I_B + 4.8 \times 0.271 = -0.7 + 3$$

$$\Rightarrow I_B = 0.002 \text{ mA}$$



$$\therefore I_C = I_E - I_B$$

$$= 0.271 - 0.002 \text{ mA}$$

$$= 0.269 \text{ mA}$$

KVL on L_2 ,

$$4.8 I_E = 3 - V_{CE} + 3$$

$$\Rightarrow V_{CE} = 6 - (4.8 \times 0.271)$$

$$\Rightarrow V_{CE} = 4.7 > 0.2$$

$\therefore$ assumption correct!

Easier to just:

$$V_{CE} = 3 - (-1.7) > 0.2$$

$$\therefore \beta = \frac{I_C}{I_B} = \frac{0.269}{0.002} = 134.5$$

$$\alpha = \frac{I_C}{I_E} = \frac{0.269}{0.271} = 0.99$$

Answer to the Q. NO-7

(a) let's assume active mode,

$$\beta = 100, V_{BE} = 0.7 \text{ V}$$

$$\alpha = 0.99, V_{CE} > 0.2$$

$$\text{Here, } V_B = 0 \text{ V}$$

$$\therefore V_E = 0 - 0.7 = -0.7 \text{ V}$$

We know,

$$I_E = I_B + I_C$$

$$\Rightarrow I_B + I_C - I_E = 0 \quad \text{--- (i)}$$

$$I_C = \beta I_B$$

$$\Rightarrow 100 I_B - I_C = 0 \quad \text{--- (ii)}$$

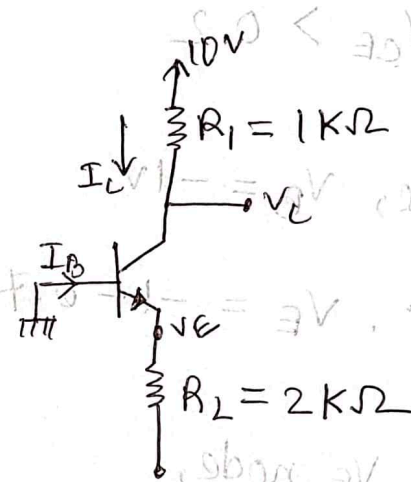
$$I_C = \alpha I_E$$

$$\Rightarrow I_C - 0.99 I_E = 0 \quad \text{--- (iii)}$$

$$A_{m \text{ S.F.}} = \beta I_B$$

$$V_{CE} = 1.05 \text{ V}$$

(Ans)



Solving eqn (i), (ii), (iii) we get,

$$I_B = I_C = I_E = 0 \text{ mA}$$

$$\therefore V_C = 10 \text{ V}$$

$$\therefore V_{CE} = 10 + 0.7 = 10.7 > 0.2 \quad [\text{active mode}]$$

$\therefore$ assumption correct!

$$I_C = I_B = I_E = 0 \text{ mA}$$

$$\text{and } V_C = 10 \text{ V} \quad (\text{Ans})$$

(b) Here,

$$V_{GS} = 10 \text{ V}$$

$$V_S = 0 \text{ V}$$

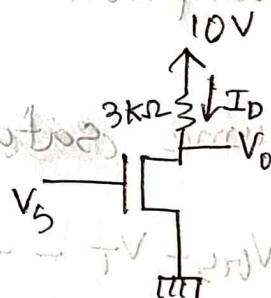
$$V_T = 1 \text{ V}$$

$$K = 5 \text{ mA/V}^2$$

$$\therefore V_{GS} = 10 - 0 = 10 > V_T$$

let's assume saturation mode,

$$I_D = \frac{K}{2} (V_{GS} - V_T)^2$$



$$\Rightarrow I_D = \frac{5}{2} (10 - 1)^2$$

$$\Rightarrow I_D = 202.5 \text{ mA}$$

At node V_0 ,

$$\frac{10 - V_0}{3} = 202.5$$

$$\Rightarrow V_0 = -507.5 = V_D$$

$$\therefore V_{DS} = -507.5 \nless V_{ov}$$

$\therefore$ assumption wrong.

let's assume active mode,

$$I_D = K(V_{GS} - V_T - \frac{1}{2} V_0) V_0$$

$$\Rightarrow \frac{10 - V_0}{3} = 5(10 - 1 - \frac{1}{2} V_0) V_0 = 5V_0(9 - \frac{1}{2} V_0)$$

$$\Rightarrow \frac{10 - V_0}{3} = 45V_0 - \frac{5}{2} V_0^2$$

$$\Rightarrow 10 - V_0 = 135V_0 - 7.5V_0^2$$

$$\Rightarrow 7.5V_0^2 - 136V_0 + 10 = 0$$

$$\therefore V_0 = \frac{18.06, 0.074}{X}$$

$$V_{ov} = 10 - 1$$

$$A_{v0} = g_m V = g_m I = g_m I$$

$$V_{ov} = 10V$$

$$f(0) = f(0 + 0) = 20V$$

assumption correct!

$$A_{v0} = g_m I = g_m I = g_m I$$

(2mA)

$$V_{ov} = 10V$$

(d) Here

$$V_{ov} = 10V$$

$$K = 20 \text{ mA/V}^2$$

$$\therefore V_{DS} = 0.074 < V_{ov}$$

$\therefore$ assumption correct!

$$f(V_{ov} - 20V) \cdot \frac{K}{2} = 0$$

$$\therefore I_{D_{S_2}} = \frac{10 - 0.074}{3} = \cancel{3.309 \text{ mA}} \quad 3.309 \text{ mA}$$

$$\text{and } V_0 = 0.074 \text{ V} \quad (\text{Ans})$$